Jeonggyu Hwang

50 Yonsei-ro, Seodaemun-gu, Seoul, Republic of Korea (03722)
+82) 10 - 2766 - 0335 | jeonggyuhwang04@gmail.com | https://developer-jg.github.io/

RESEARCH INTERESTS

- Phase-Change Nanophotonics
- Hyperbolic Metamaterials
- Optical Hyperlenses
- Phase-Transition Oxides
- Computational Nanophotonics

EDUCATION

Yonsei University **Seodaemun-gu, Seoul, South Korea**
Combined Master-Doctor Student March 2026 - Present
School of Electrical & Electronic Engineering, College of Engineering

Gachon University **Seongnam-si, Gyeonggi-do, South Korea**
Undergraduate Student March 2023 - February 2026
Major in Semiconductor Display, Department of Semiconductor Engineering, College of Semiconductor
- Bachelor of Engineering (B.E.)
- GPA (Overall): 4.33 / 4.50 (Major GPA: 4.37 / 4.50), Graduation with Honors

Seoul National University **Gwanak-gu, Seoul, South Korea**
Undergraduate Exchange Student August 2024 - February 2025
Department of Materials Science and Engineering, College of Engineering

PUBLICATIONS

Conference:

[3] Hwang, J., & Lee, T. (2026). Chemical Bonding Engineering of GeS_2 Ovonic Threshold Selectors via Nitrogen Incorporation. Poster presentation at MRS-K Fall Meeting

[2] Hwang, J., & Lee, T. (2026). Study of germanium sulfide films with different nitrogen doping levels for ovonic threshold switches. Poster presentation at GCIM 2026 & MRS-K Spring Meeting.

[1] Hwang, J., & Jeon, J. (2025). Exploring the theoretical limits of efficiency in multilayer solar cells. Poster presented at the 32nd Korean Conference on Semiconductors (KCS), Korea.

Pre-Print:

[3] Hwang, J. (2025). Estimation of Solar Spectral Irradiance Using Meteorological Data and Analysis of Optimal Conditions for Solar Power Generation. *arXiv preprint arXiv:2504.08008*.

[2] Hwang, J. (2024). Theoretical analysis of doping concentration gradients on solar cell performance. *arXiv preprint arXiv:2407.12245*.

[1] Hwang, J. (2024). Exploring the theoretical limits of efficiency in multilayer solar cells. *arXiv preprint arXiv:2404.14930*.

ACADEMIC EXPERIENCE

Yonsei University **Seodaemun-gu, Seoul, South Korea**
Graduate Researcher January 2026 - Present
School of Electrical & Electronic Engineering, College of Engineering
- NanoBio Device Lab. (NBDL)

- PI: Prof. Lee, Taeyoon
- Research topic: TBD

Undergraduate Researcher June 2025 - July 2025
Department of Physics, College of Science
- Center for Bandstructure Engineering
- PI: Prof. Kim, Keunsu
- Research topic: 2D Materials and Solid-State Physics

POSTECH **Pohang-si, Gyeongsangbuk-do, South Korea**
Pohang University of Science and Technology
Undergraduate Researcher February 2024
Department of Chemical Engineering
- FIMDEM Lab. - Functional Inorganic Materials Design & Manufacturing
- PI: Prof. Son, Jaesung
- Research topic: Solution-Processed Thin Films and Thermoelectric Device Materials

Korea University **Seongbuk-gu, Seoul, South Korea**
Undergraduate Researcher July 2023 - August 2023
Department of Integrative Energy Engineering, College of Engineering
Graduate School of Energy and Environment
- Electrochemistry Lab. - Solarrino
- PI: Prof. Jun, Yongseok
- Research topic: Perovskite Solar Cells

INDUSTRY EXPERIENCE

Philoptics Co., Ltd **Osan-si, Gyeonggi-do, South Korea**
Engineer March 2024 - May 2025
Semiconductor Department, R&D Division
- Development of Advanced Glass Singulation Processing Technology
- Development of Next-Generation ABF Drilling Core Technology
- 2024 Advanced Technology Development Project

AWARDS AND HONORS

Incoming Excellence Scholarship, Yonsei University
Graduation with Honors, Gachon University
Selected for Priority Admission, Yonsei University
Academic Excellence Scholarship, Gachon University
Undergraduate Poster Presentation Award,
32nd Korean Conference on Semiconductor

LEADERSHIP & ACTIVITY

Research Service & Leadership
The Gachon Research Society
Honorary Member February 2025 - Ongoing
Chairperson of the Acting Executive & Steering Committee for the 2025 Delegates Joint Meeting February 2025 - November 2025
Executive Adviser December 2024 - February 2025
Founding President & 1st President February 2024 - November 2024
- Contributed to improving language proficiency and fostering academic development

through reading and debating multilingual papers and books.
- Facilitated information exchange within the university and strengthened networking opportunities among student researchers.

Academic Service & Leadership

Teaching Assistant (TA)

Physical Electronics (EEE6807), Fall 2026 September 2026 - December 2026
School of Electrical & Electronic Engineering, Yonsei University

Physical Electronics (EEE3210), Spring 2026 March 2026 - June 2026
School of Electrical & Electronic Engineering, Yonsei University

Administrative Assistant

School of Electrical & Electronic Engineering, Yonsei University January 2026 - Present

Executive Member

Student Association March 2023 - February 2024
Major in Semiconductor Display, Gachon University

Jeonggyu Hwang

Graduate Student

EDUCATION

March 2025 - Present
Combined Master-Doctor Student in
School of Electrical & Electronic Engineering, Yonsei University
> Supervisor: Prof. Lee, Taeyoon Ph.D.
> Research topic: Phase-Change Hyperlens

RESEARCH EXPERIENCE

January 2025 - Present
Graduate Research Assistant
NanoBio Device Lab. (NBDL), Yonsei University

ACADEMIC EXPERIENCE

September 2026 - December 2026
Teaching Assistant (TA)
Physical Electronics (EEE6807), Fall 2026

March 2026 - June 2026
Teaching Assistant (TA)
Physical Electronics (EEE3210), Spring 2026

January 2026 - June 2026
Administrative Assistant
School of Electrical & Electronic Engineering, Yonsei University

PUBLICATIONS

Conference

[2] **Hwang, J.**, & Lee, T. (2026). Chemical Bonding Engineering of GeS_2 Ovonic Threshold Selectors via Nitrogen Incorporation. Poster presentation at MRS-K Fall Meeting.

[1] **Hwang, J.**, & Lee, T. (2026). Study of Germanium Sulfide films with Different Nitrogen Doping Level for OvonicThreshold Switch. Poster presentation at GCIM 2026 & MRS-K Spring Meeting.

HONORS & AWARDS

November 2025
April 2026
Priority Selection for Yonsei Graduate school
Incoming Excellence Scholarship for School of Electrical & Electronic Engineering, Yonsei University

PERSONAL INFO

Address
50 Yonsei-ro, Seodaemun-gu, Seoul, Republic of Korea (03722)

Phone
+82) 10 - 2766 - 0335

E-mail
jeonggyuhwang04@gmail.com

Portfolio
https://developer-jg.github.io/

Last Updated : October 20, 2026